



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2024

Time: 3 Hrs.

Marks: 60

Subject: Parametric and Nonparametric Tests (Theory)
Course Code: STAT-301

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(5x6=30)

- A manufacturer of car batteries guarantees that the batteries will last, on average, 3 years with a standard deviation of 1 year. If five of these batteries have lifetimes of 1.9, 2.4, 3.0, 3.5, and 4.2 years, should the manufacturer still be convinced that the batteries have a standard deviation of 1 year at 5% level of significance?
- Clearly differentiate between one-tailed and two-tailed tests. Support your answer by giving suitable examples.
- Write complete procedure for Bartlett's test for equality of more than two variances, also give its underlying assumptions.
- Define the term sequential analysis and write general procedure of sequential test.
- The 95% confidence interval calculated for mean of a normal distribution having standard deviation 120 is (1023.3, 1101.7). Test the hypothesis that population mean is 1100 at 1% level of significance.

Q.2. Answer the following questions.

(3x10=30)

Following table shows how many times busses were late arriving at the respective stops in 100 weeks. Test whether binomial distribution is a good fit to the data.

No. of times busses were late	0	1	2	3	4	5	6	7
No. of weeks	0	2	8	19	29	26	13	3

A random sample of size 16 yielded mean 24.48 and variance 2.25, another sample of size 25 yielded mean 27.50 and variance 1.96. Test the hypothesis at 5% level of significance that:

- Both samples are taken from populations having same variances.
- Difference between two population means is more than 2.

A shoe company wants to know if three groups of workers have different salaries (000 PKR):

Women	52, 50, 50, 51, 52, 52, 51, 52, 52,
Men	53, 50, 52, 52, 50, 49, 50, 51, 52, 50, 50, 51, 50,
Minorities	43, 42, 44, 42, 42, 42, 43, 42, 43, 42,

Use Kruskal Wallis H test to analyze whether there is a significant difference in the salaries of these groups working for this company.